Short Communication

Study of TATP: Spontaneous Transformation of TATP to DADP

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Received: August 1, 2007; revised version: October 10, 2007

DOI: 10.1002/prep.200700247

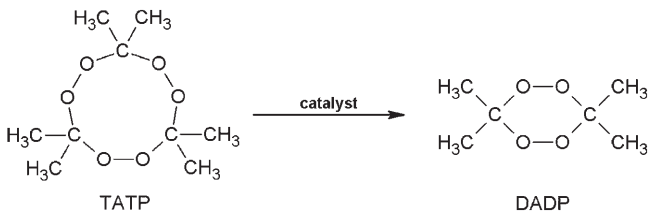
Abstract

TATP, prepared in the presence of catalysts methanesulfonic, perchloric, or sulfuric acid, has been found to undergo transformation to DADP. However, no transformation occurs if TATP is purified or prepared involving catalysts such as hydrochloric acid, tin(IV) chloride, and nitric acid. The transformation has been monitored by the methods of DTA and HPLC.

Keywords: DADP, DTA, HPLC, TATP, Transformation

1 Introduction

The reaction of acetone and hydrogen peroxide in the presence of an acid catalyst under common conditions yields a cyclic trimer – TATP (3,3,6,6,9,9-hexamethyl-1,2,4,5,7,8-hexoxonane) – with a small amount of side product reported to be a tetramer – TeATeP (3,3,6,6,9,9,12,12-octamethyl-1,2,4,5,7,8,10,11-octaoxacyclododecane) – by Schulte-Ladbeck et al. [1], Pena et al. [2] or a structural conformer of TATP by Widmer et al. [3]. However, under certain reaction condition, the presence of cyclic dimer – DADP (3,3,6,6-tetramethyl-1,2,4,5-tetroxane) has also been observed [2, 4, 5]. In this short communication, we report the spontaneous transformation of TATP to DADP (Scheme 1).



Scheme 1. Schematic representation of transformation of TATP to DADP.

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2 Results and Discussion

The reaction of acetone and hydrogen peroxide, in the presence of hydrochloric acid, sulfuric acid, nitric acid, methanesulfonic acid, perchloric acid, or tin(IV) chloride as catalyst yields TATP as the main product [6]. Up to a molar ratio of catalyst to acetone $n_c/n_a = 0.5$, it is contaminated with a small amount of side product irrespective of the catalyst used. This side product, reported to be TeATeP [1, 2] or a structural conformer of TATP [3], was observed but not studied in the scope of this work. The DADP was not found in any of the samples prepared up to this molar ratio of catalyst to acetone. The type of catalyst did not influence the composition of the product. Although the composition was the same, the stability of the product (washed to neutrality but not recrystallized) varied significantly depending on the type of catalyst used during preparation [6, 7].

During our studies of thermal stability of TATP [7], we found that some preparation routes yield TATP which if left alone transforms to DADP. This transformation was first observed by differential thermal analysis (DTA) as a complete change of thermoanalytical behavior (Figure 1) and more recently confirmed by HPLC (Figure 2 demonstrates this behavior for TATP stored at 0 °C).

Whether TATP transforms to DADP or not depends on the type of catalyst used during its preparation. Recrystallized TATP as well as TATP prepared with the use of hydrochloric acid or tin(IV) chloride does not undergo this transformation, trace amounts of DADP were found after 2 months when nitric acid was used. On the other hand, TATP produced by the same procedure but catalyzed by methanesulfonic, perchloric, or sulfuric acid transforms to DADP. The transformation is quite rapid at 60 °C but has been observed to take place even at 0 °C as demonstrated in Figure 2. The transformation rate depends on the storage temperature and the amount of catalyst used in the preparation of TATP.

Pure TATP, with m.p. in the range of 95 – 98.5 °C and decomposition above 150 – 160 °C [4, 8 – 12], does not

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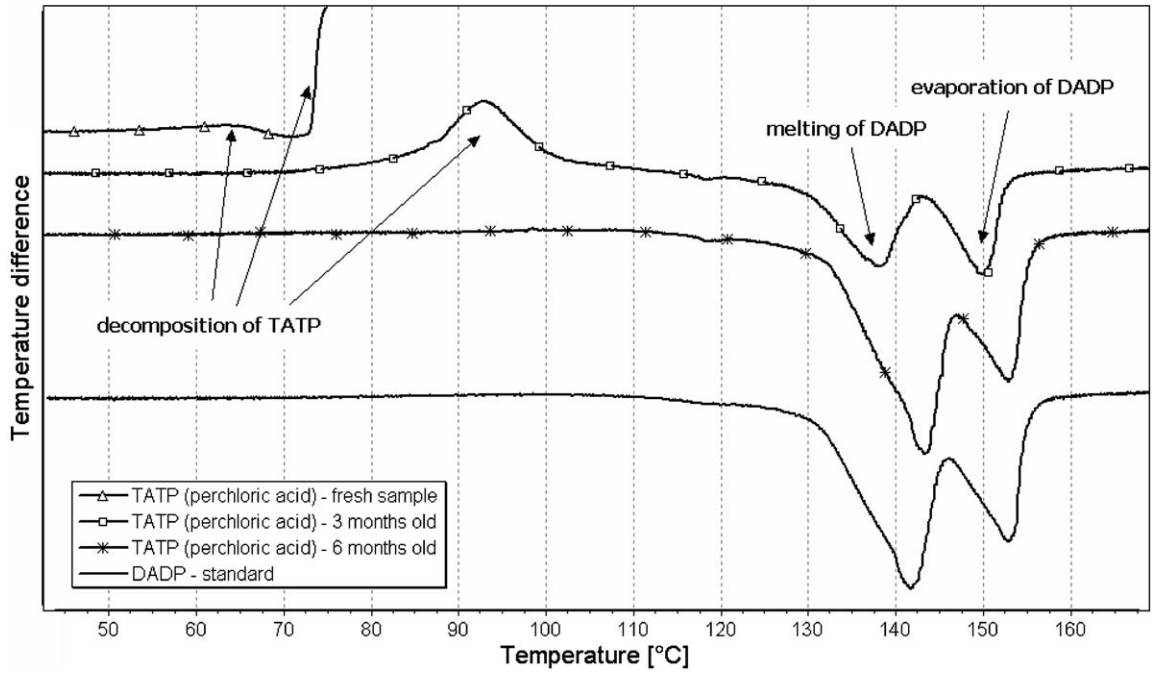


Figure 1. DTA thermogram of fresh TATP from perchloric acid ($n_c/n_a = 0.25$) [7], the same TATP after 3 and 6 months storage in laboratory and pure DADP for comparison (heating rate 5 °C min^{-1}, 30 mg samples, static air atmosphere, open test tubes, thermograms shifted along y axis).

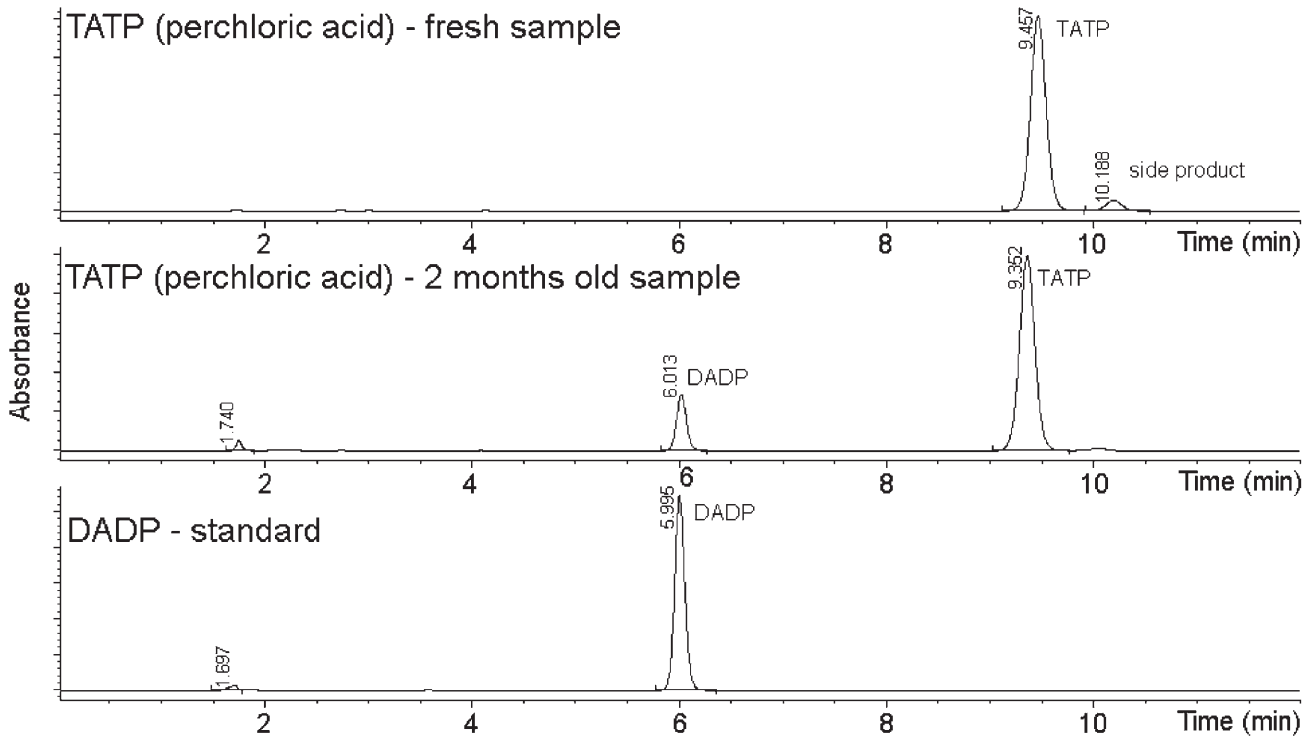


Figure 2. HPLC chromatogram of fresh TATP from perchloric acid ($n_c/n_a = 0.25$) [7], the same TATP after 2 months of storage at 0 °C and pure DADP for comparison.

undergo spontaneous transformation, and for the sake of clarity is not shown in Figure 1. The thermoanalytical behavior of TATP prepared from perchloric acid depends on n_c/n_a and may be significantly different from that observed for pure TATP [7]. When freshly prepared with $n_c/n_a = 0.25$, it decomposes before melting as displayed in Figure 1.

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The DTA thermogram in Figure 1 clearly demonstrates the thermoanalytical behavior of aging TATP, prepared using perchloric acid as catalyst. The amount of DADP steadily increases on standing at 25 °C until full conversion is achieved. The transformation from TATP to DADP is most apparent from the gradual shift of the beginning of decomposition to the higher temperatures and decrease in the peak's height followed by appearance of double endothermal peaks above 120 °C which is typical for pure DADP.

The chromatographic method used for confirmation of transformation behavior gives clear baseline separation as can be seen in Figure 2. The retention time of DADP is 6.0 min and for TATP 9.4 min. The small peak which appears at 10.2 min in the chromatogram is probably due to the tetrameric TeATeP [1] or the TATP conformer [3]. This peak disappears during aging.

3 Conclusion

TATP prepared by the direct reaction of acetone and hydrogen peroxide in the presence of catalysts such as hydrochloric acid, tin(IV) chloride, or nitric acid has been found to be stable. However, in the presence of catalysts such as methanesulfonic, perchloric, or sulfuric acid, TATP isolated has been found to undergo spontaneous gradual transformation to DADP on standing. The amount of catalyst used and the temperature of storage have an influence on the rate of the transformation.

4 References

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Acknowledgements

We would like to acknowledge Dr. Miloslav Krupka from OZM Research for providing prototype of DTA Ex 550 for experiments with primary explosives.

Symbols and Abbreviations

n_c/n_a Molar ratio of acid catalyst to acetone